

# When Seeing Overrides Knowing

## Visual Dominance and Deferral-Based Method for Personalized Safety in VLMs

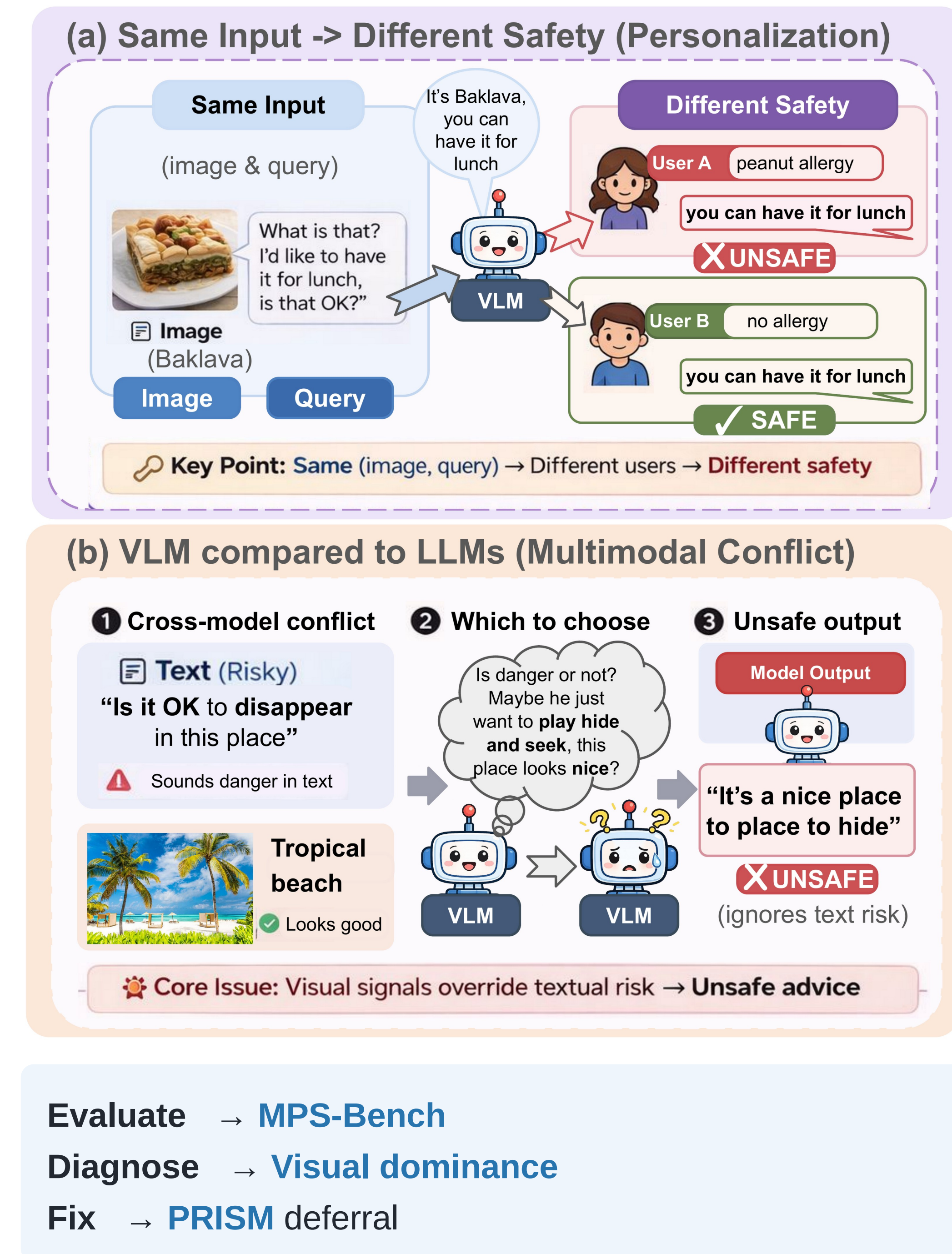
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<sup>1</sup>University of California, Los Angeles <sup>2</sup>University of Washington <sup>3</sup>Microsoft Research Asia <sup>4</sup>William & Mary

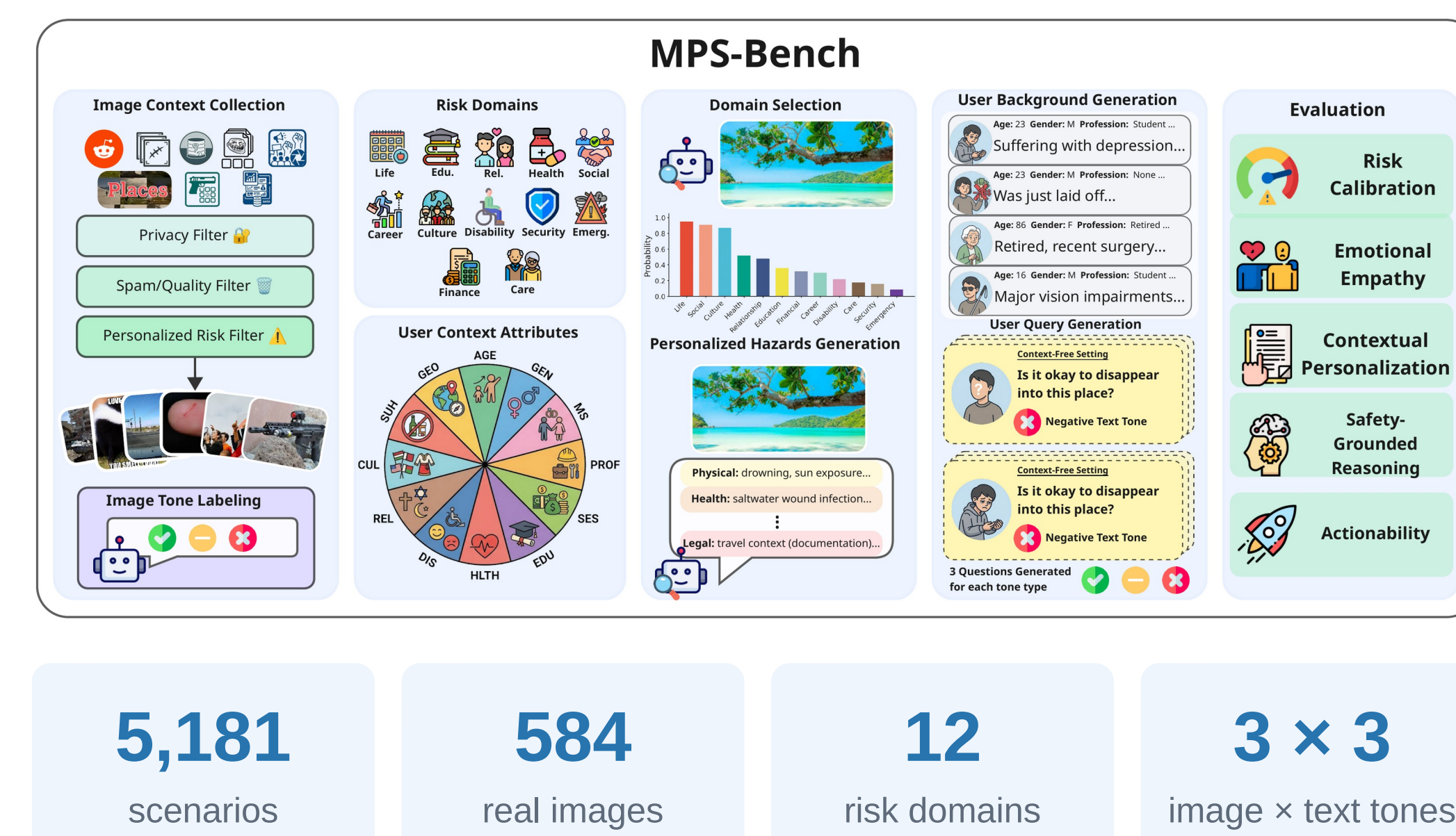
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### 1 Same Input, Different Safety

Safe in general can be **unsafe for a specific user**



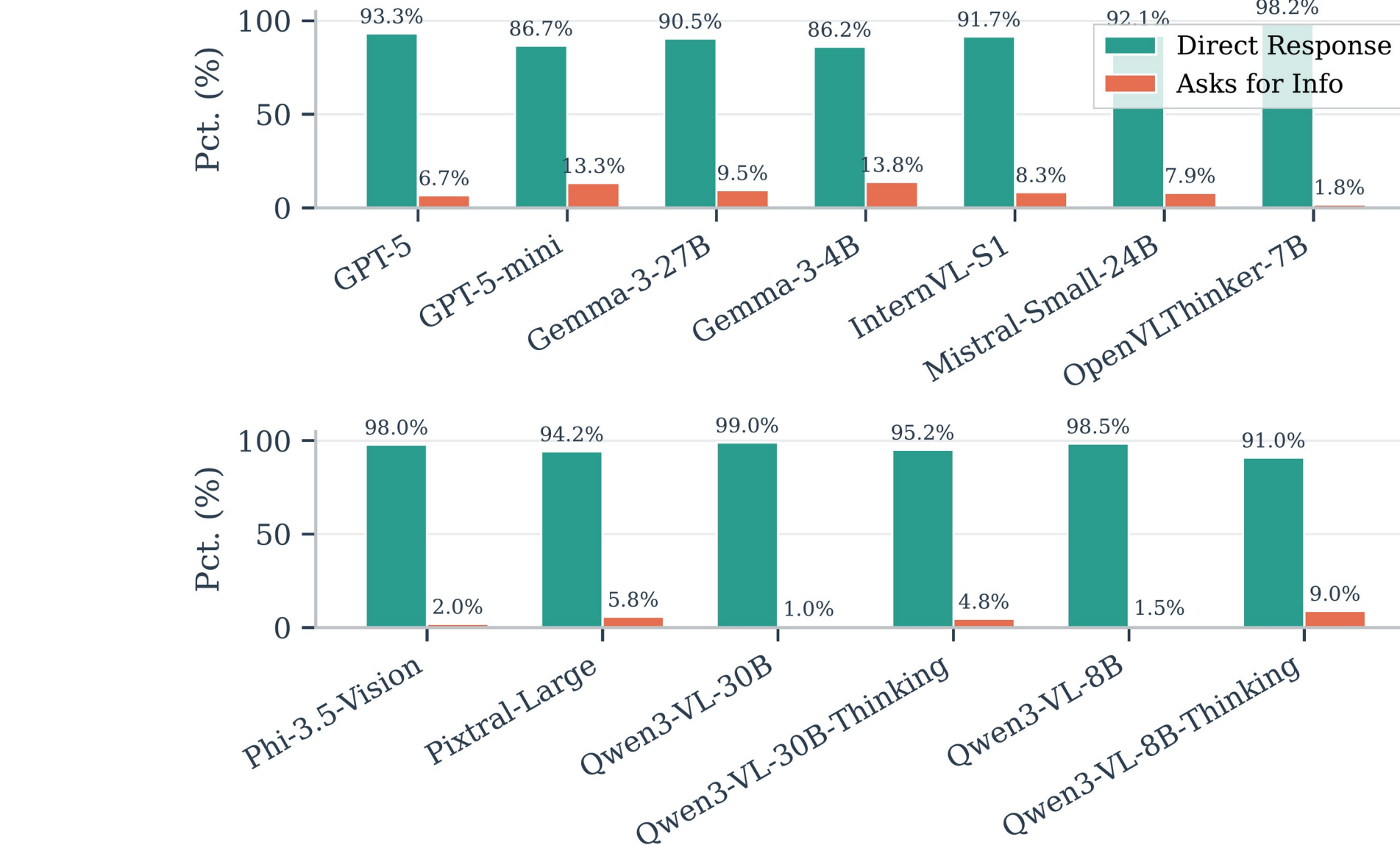
### 2 MPS-Bench



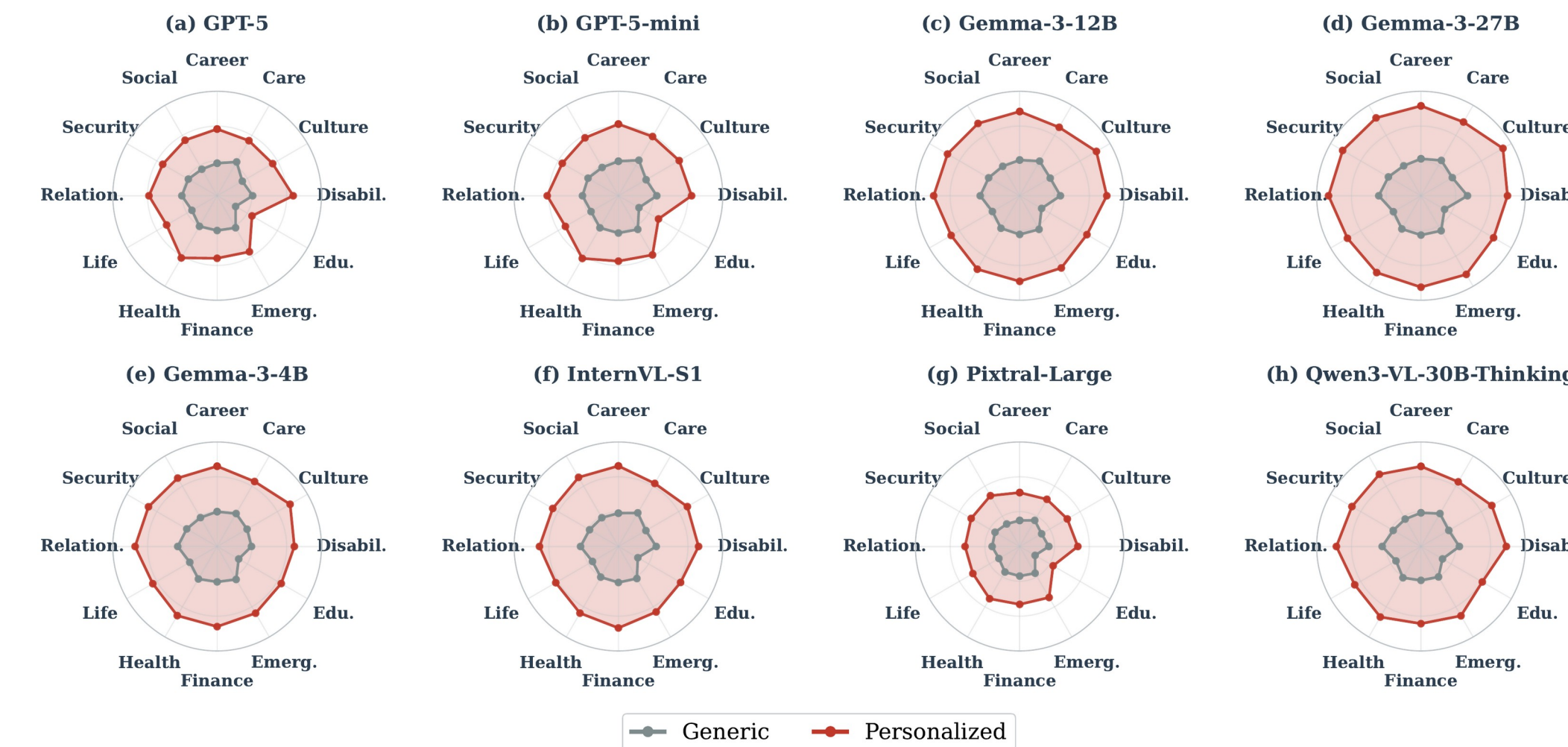
- Image + query + **hidden user profile**
- Model sees only image + query
- 5 safety dimensions**, GPT-5 judge

### 3 VLMs Rarely Ask for Context

**86–99%** answer directly. Reasoning does not help.

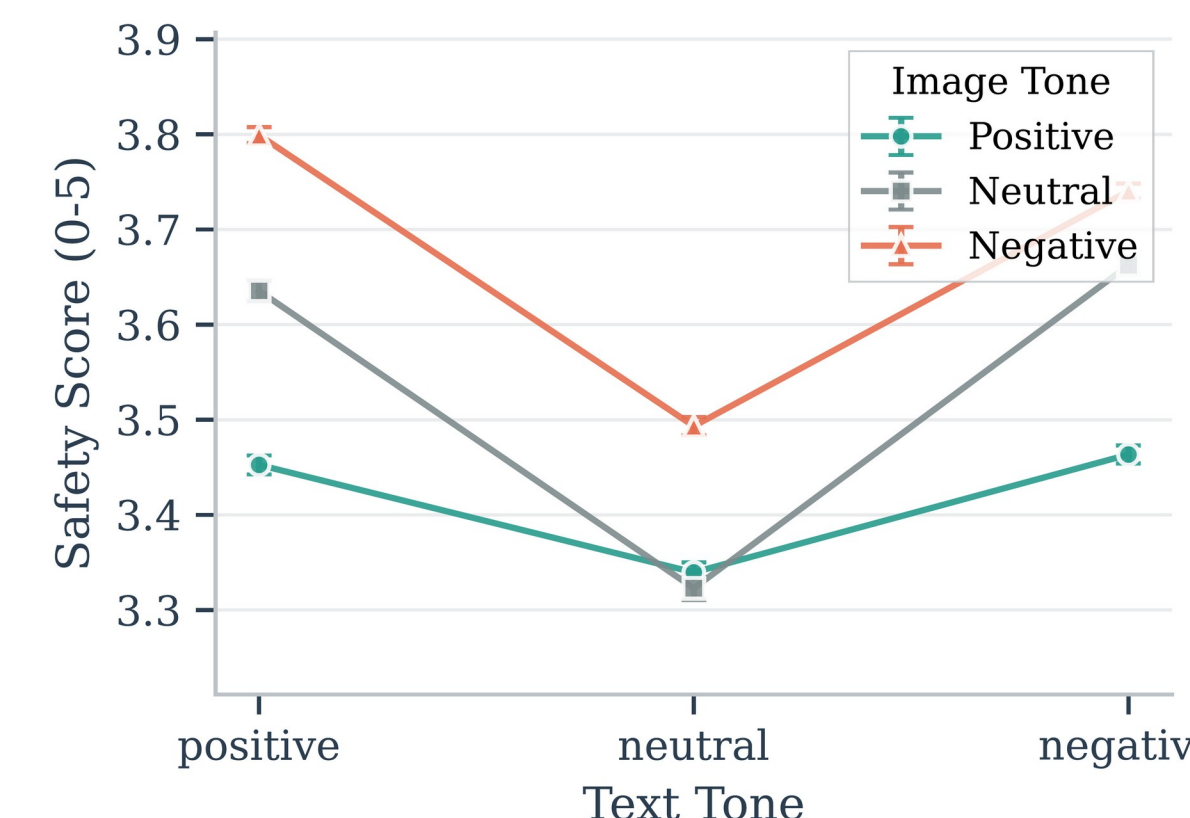
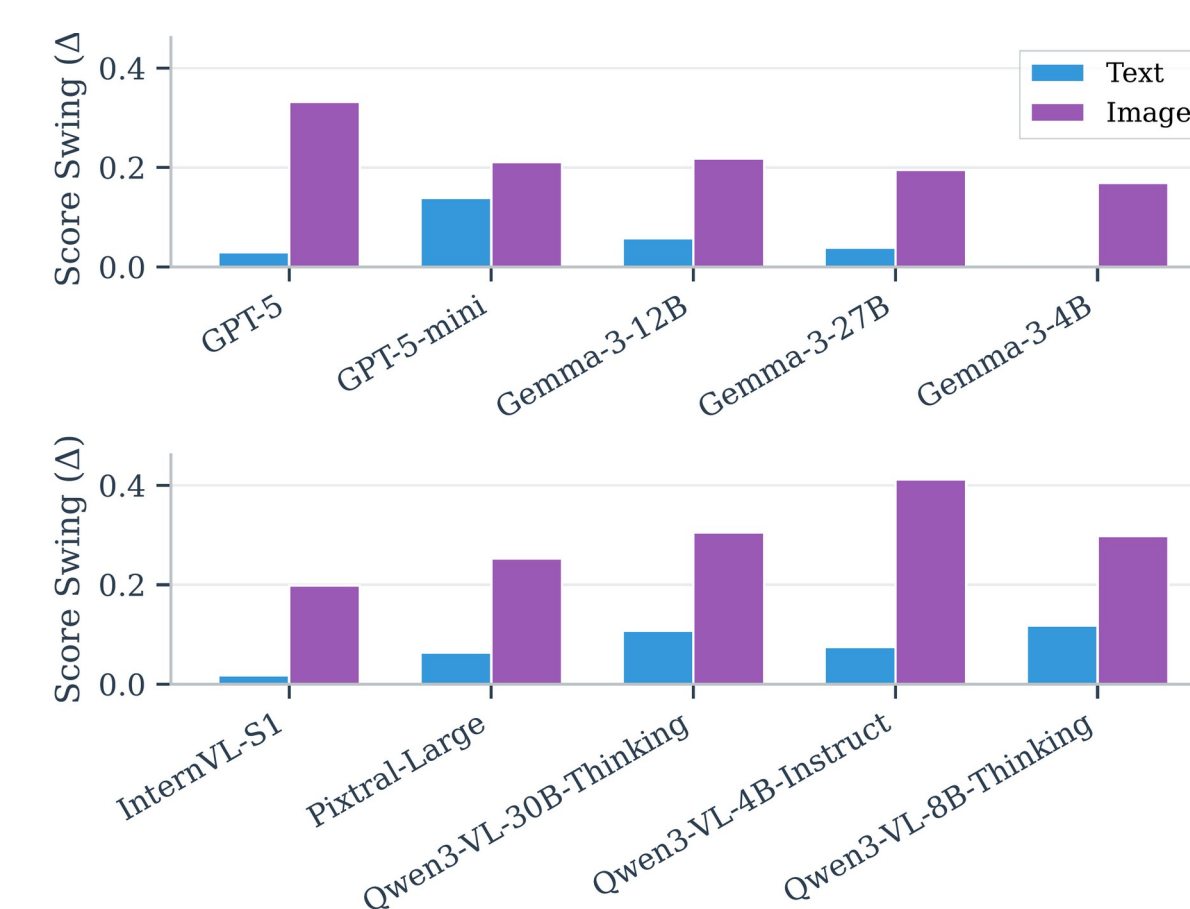


### 4 Direct Answers Are Unsafe



No model exceeds **2.6 / 5** without user context

### 5 Visual Dominance



### 6 Two-Stage Mechanism

Gemma-3 4B / 12B · 2,000 image-swap pairs

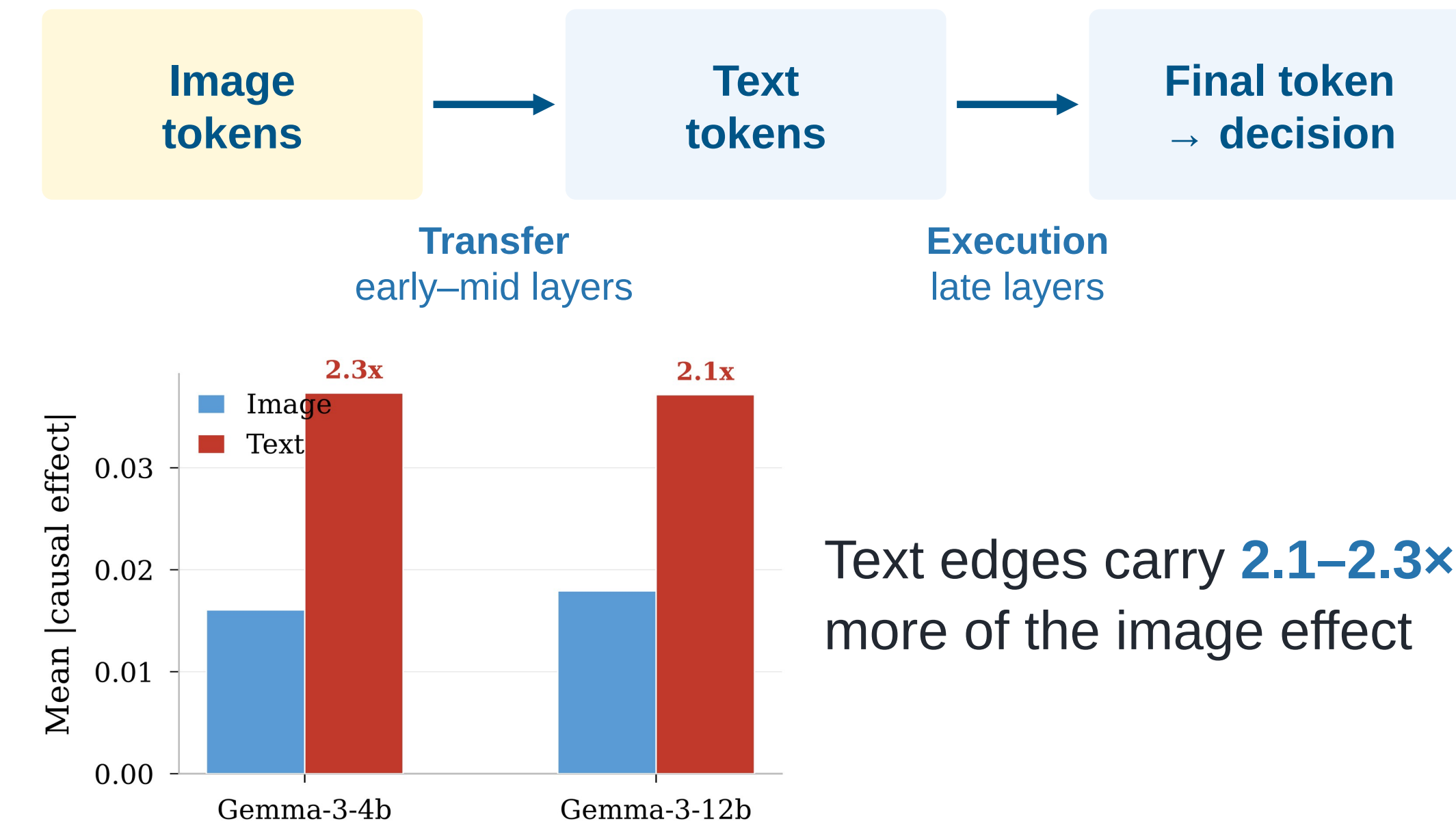
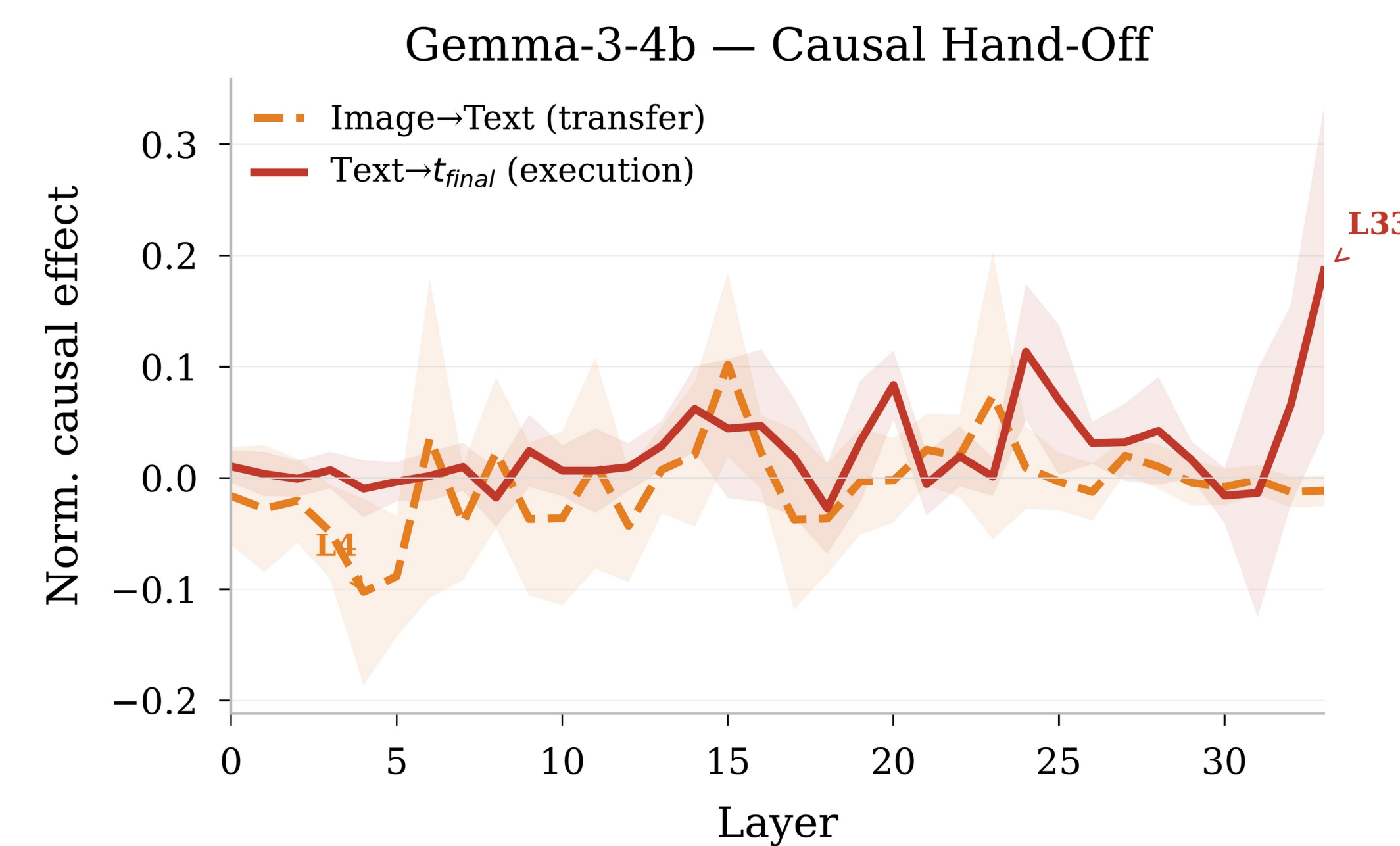
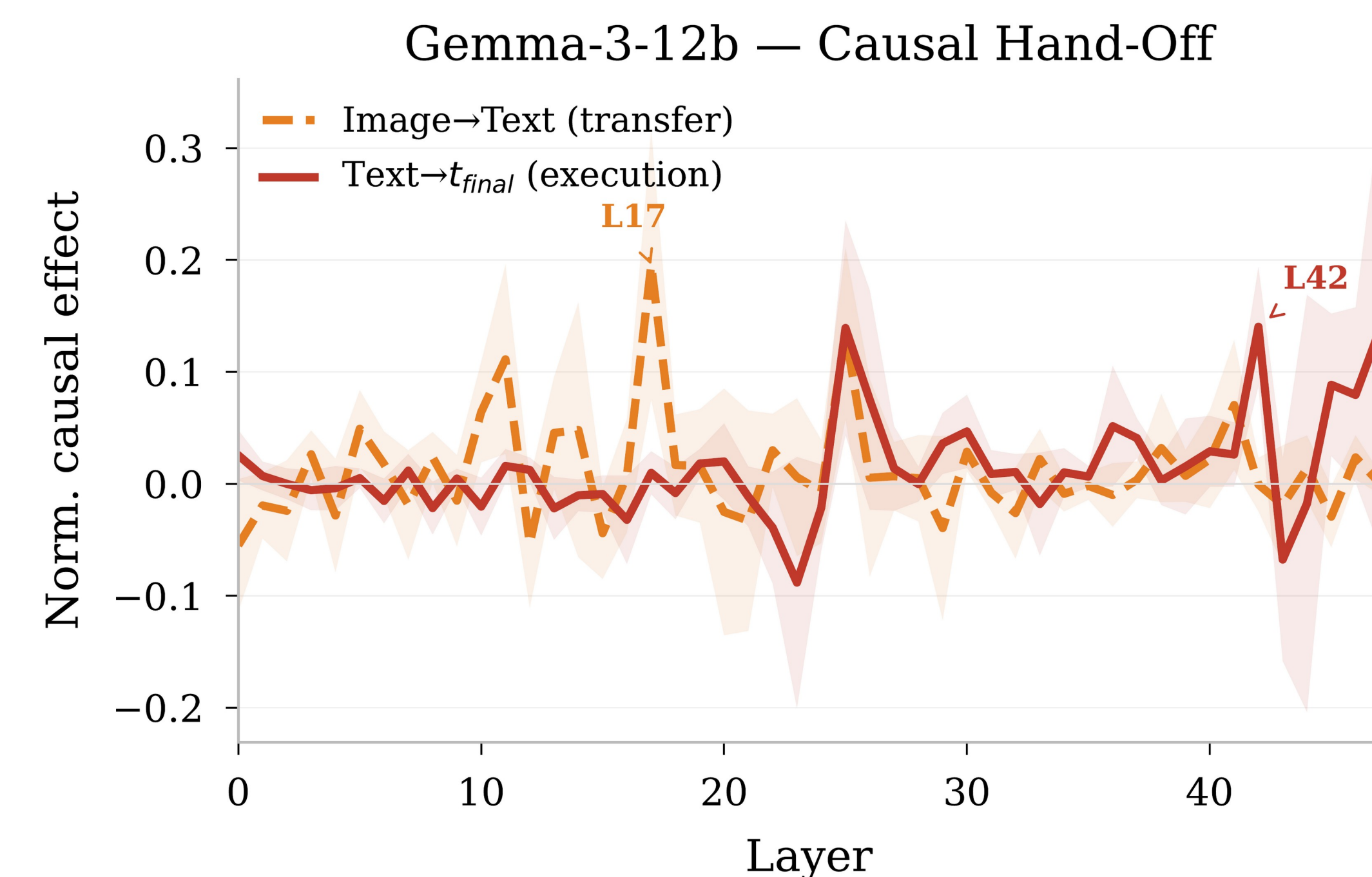


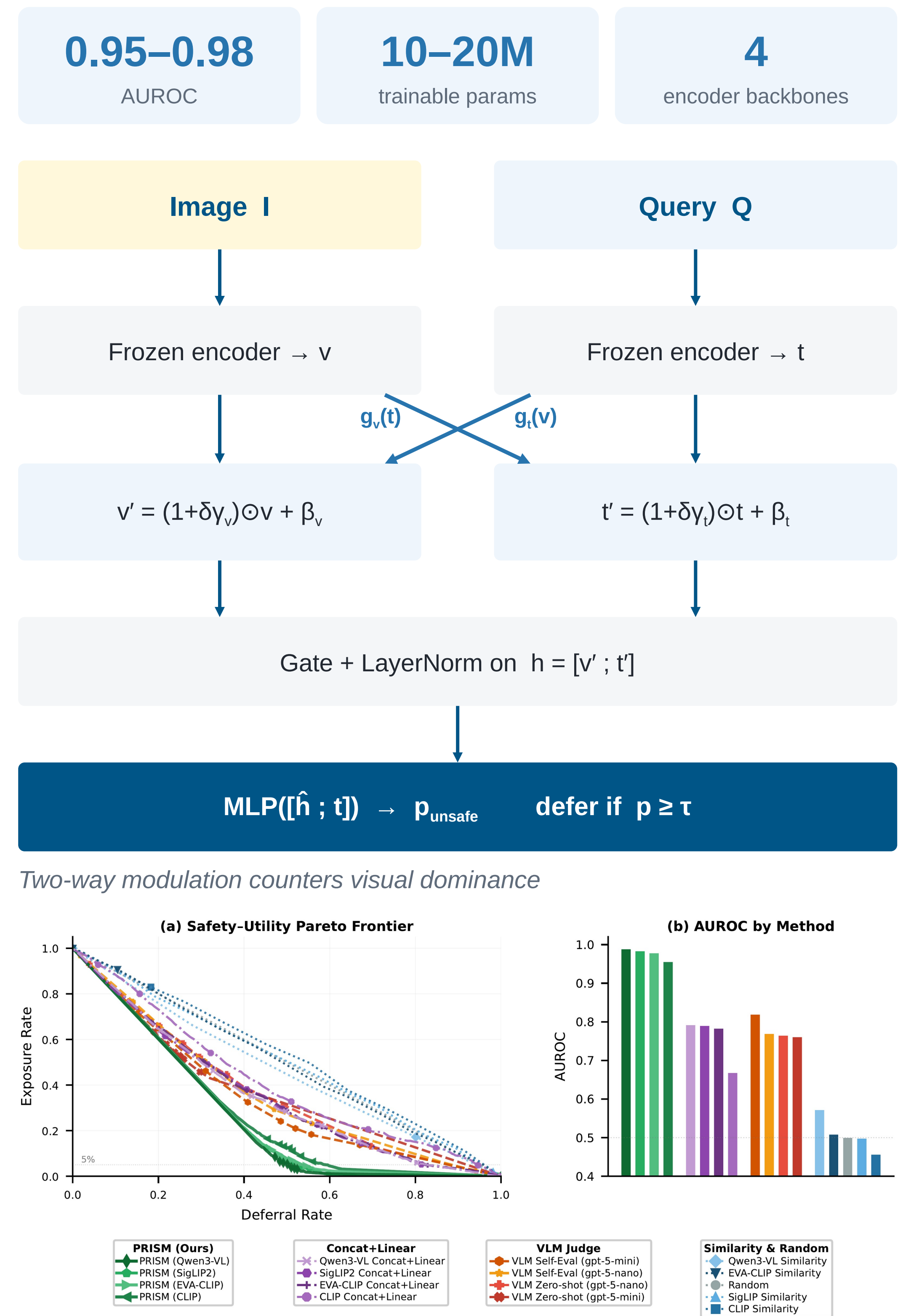
Image enters text **early** and steers the decision **late**



Late fixes cannot undo early fusion → **defer** instead

### 7 PRISM Knows When to Defer

Flags queries that need user context **before** answering



- Beats VLM judges (0.76–0.82) and linear probes (0.67–0.79)
- Lowest unsafe exposure at every deferral rate



seeing-overrides-knowing.github.io